NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – II : SOLAR ENERGY COLLECTION

1. Introduction – Solar Collectors

A solar collector is a special heat exchanger that absorbs solar radiation, converts it to heat and transfers the heat to a working fluid (water, air, oil).

❖Classification

- Non-concentrating (stationary): aperture area = absorber area → flat plate collector (FPC), evacuated tube collector (ETC). Gives low-medium temperature (40–100 °C).
- Concentrating (focusing): aperture area > absorber area → reflectors/ lenses focus rays. Gives 100–2000 °C.
- By fluid: liquid collectors (water, oil), air collectors.
- By tracking: non-tracking, single-axis, two-axis.
- Concentration ratio $C = A_a / A_r$ (aperture area / receiver area).

2. Flat Plate Collector (FPC)

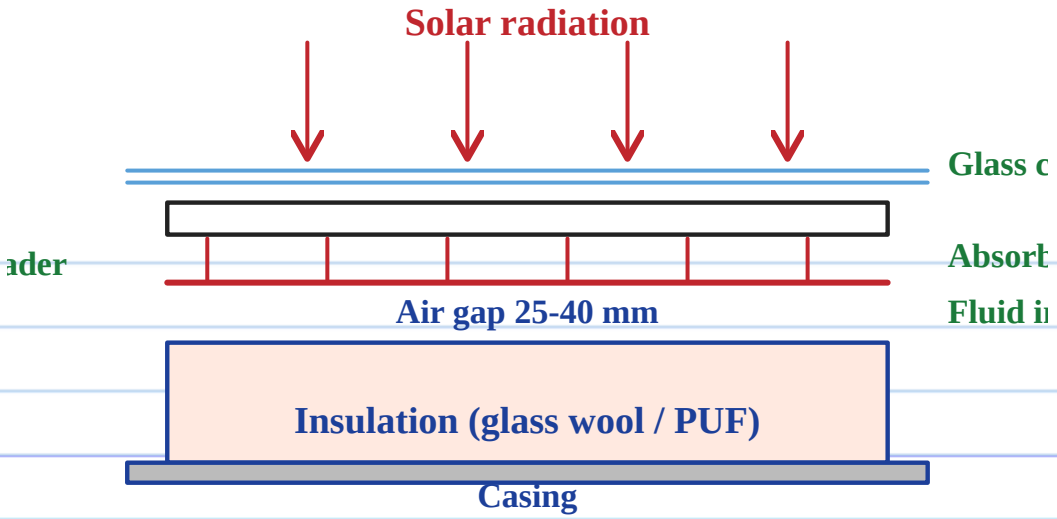


Fig 2.1 Cross-section of a liquid flat plate collector

❖Components

- Absorber plate: copper / aluminium / steel sheet, blackened with selective coating (e.g., black chrome, black nickel) – high absorptance α and low emittance ε.

- **Tubes (risers) and headers:** tubes welded/brazed to the plate carry the fluid; pitch 10–15 cm, tube dia 1–1.5 cm.
- **Transparent cover:** one or two glass sheets (3–5 mm) – transmits short wave radiation, **traps long-wave re-radiation** (greenhouse effect); also cuts convection loss.
- **Insulation:** glass wool / mineral wool / PUF, 5–10 cm at back and sides.
- **Casing (box):** aluminium/GI/FRP; weather-proof.

❖Working

- Solar radiation passes through glass, strikes the black absorber plate and is converted to heat; heat flows to the tubes and into the fluid by conduction and convection.
- Thermo-siphon (natural circulation) or pump-driven circulation to a storage tank.
- Tilt = latitude; faces south; efficiency 40–60%; fluid temp up to ~ 80–90 °C; stagnation ~ 150 °C.

❖Advantages & limitations

- Absorbs **both beam & diffuse** radiation; no tracking; simple, cheap, low maintenance.
- Limitations: low temperature, large heat loss at high ΔT, efficiency drops on cloudy days.

❖Selective surface

> **Selective surface:** high α for short-wave (0.3–3 μm), low ε for long-wave (> 3 μm) so it absorbs well yet radiates poorly. Ratio $\alpha/\varepsilon > 10$ (e.g. black chrome $\alpha = 0.95$, $\varepsilon = 0.1$).

3. Thermal Analysis of Flat Plate Collector

❖Energy balance

- Radiation absorbed (per unit area) $S = I_T (\tau\alpha)$ where τ = transmittance of cover, α = absorptance of plate, I_T = radiation on tilted surface.
- Heat loss = $U_L (T_p - T_a)$

$$\text{Useful heat gain } Q_u = A_c [S - U_L (T_p - T_a)]$$

- Also $Q_u = \dot{m}\, C_p (T_{fo} - T_{fi})$.

❖Overall loss coefficient $U_L = U_t + U_b + U_e$

- U_t top loss (convection + radiation through the glass) ≈ 3–8 W/m^2K
- U_b bottom loss = k/L of insulation ≈ 0.5 W/m^2K
- U_e edge loss – small.

❖Collector efficiency factor F' and heat removal factor F_R

- Since plate temperature varies along the flow, we relate gain to **inlet fluid temp** instead:

$$Q_u = A_c F_R [S - U_L (T_{fi} - T_a)]$$

- $F_R = \dot{m} C_p / (A_c U_L) \times [1 - \exp(-A_c U_L F' / \dot{m} C_p)]$
- F_R = (actual useful gain) / (gain if whole plate were at inlet fluid temp). $0 < F_R < 1$.

❖Instantaneous efficiency (Hottel–Whillier–Bliss equation)

$$\eta = Q_u / (A_c I_T) = F_R(\tau\alpha) - F_R U_L (T_{fi} - T_a) / I_T$$

- Plot η vs $(T_{fi} - T_a)/I_T$ → **straight line**: intercept = $F_R(\tau\alpha)$ (optical efficiency), slope = $-F_R U_L$ (loss).
- Higher intercept & lower slope → better collector. Used in **testing** (ASHRAE 93 / BIS).

❖Fin efficiency (standard fin of the tube-plate)

- Plate between tubes acts as a fin: $F = \tanh[m(W-D)/2] / [m(W-D)/2]$, $m = \sqrt{(U_L/k\delta)}$. W = tube spacing, D = tube dia, δ = plate thickness.

❖Example

> FPC area 2 m^2, I_T = 800 W/m^2, $(\tau\alpha)$ = 0.8, U_L = 6 W/m^2K, F_R = 0.85, T_{fi} = 40 °C, T_a = 25 °C.

$Q_u = 2 \times 0.85[800 \times 0.8 - 6 \times 15] = 1.7 \times 550 = 935$ W, $\eta = 935/(2 \times 800) = 58.4$ %.

4. Evacuated Tube Collector (ETC)

- Absorber tube inside a **vacuum-sealed glass tube** → **no convective loss**; uses heat-pipe or U-tube; efficient in cold, cloudy climates; up to 120–150 °C.
- Glass-glass (Dewar type): two concentric tubes, vacuum in between, selective coating on the inner tube.

- Costlier than FPC; fragile; but low $U_L \approx 1\text{–}2\ W/m^2K$.

5. Solar Air Heater

- Same as FPC but fluid is **air** (low C_p, low h → needs large area / fins / roughness / corrugated surface).
- Types: bare plate, **finned**, **corrugated**, **V-corrugated**, matrix-packed, **with artificial roughness**, double-pass.
- Uses: crop drying, space heating, seasoning of timber. No freezing / leakage / corrosion issues.

6. Concentrating Collectors

> A concentrating collector uses **reflectors or lenses** to focus beam radiation on a small **receiver**, hence high temperature and small heat loss area.

Concentration ratio $C = A_a / A_r$ (aperture area / absorber area)

- **Maximum possible C (ideal, 2-D):** $C_{max} = 1 / \sin \theta_a$ (θ_a = acceptance half-angle). For 3-D (point focus): $C_{max} = 1 / \sin^2 \theta_a$. The sun's half-angle = $0.267°$ → $C_{max} \approx 215$ (2-D) & 46,000 (3-D).
- **Acceptance angle ($2\theta_a$):** range of incidence angles within which all rays reach the receiver without tracking. Larger C ⇒ smaller acceptance angle ⇒ accurate tracking needed.
- They use **only beam radiation** (diffuse is lost) → need clear sky.

❖Advantages vs flat plate

- Higher temperature, higher efficiency at high temperature, less heat-loss area, less costly absorber material, can use thermodynamic cycles (power generation).

❖Disadvantages

- Need sun tracking, lose diffuse radiation, reflector surfaces get dirty/ need cleaning, high cost, complex.

7. Classification of Concentrating Collectors

Basis	Types
Focus	Line focus (trough) / Point focus (dish, tower)
Reflector	Reflecting (mirror) / Refracting (Fresnel lens)
Receiver	Convex, flat, cylindrical, cavity
Tracking	Single-axis / Two-axis / Non-tracking (CPC)
Contact	Imaging / Non-imaging

❖(a) Parabolic trough collector (PTC) – line focus

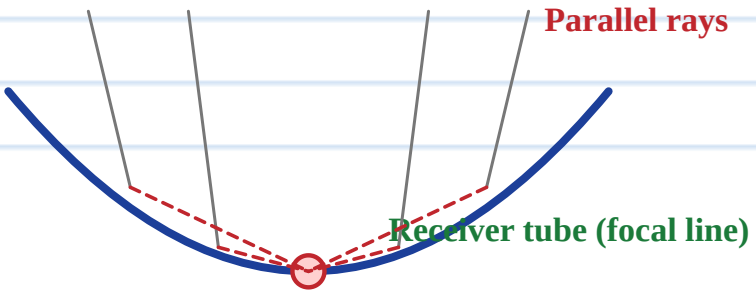


Fig 2.2 Parabolic trough concentrator (line focus)

- Parabolic cylinder mirror (aluminium/ silvered glass) with a **receiver tube** (stainless steel with selective coating inside an evacuated glass envelope) at the focal line.
- Heat transfer fluid: synthetic oil (Therminol) → 390 °C; steam generation (DSG).
- **One-axis tracking** (E–W or N–S axis). C = 15–80. Used in SEGS plants in California (354 MW) and Indian plants at Gurgaon/Rajasthan.
- **Parabola equation:** $y^2 = 4 f x$; focal length f. Rim angle ψ_r: $f / a = (1 + \cos \psi_r) / (4 \sin \psi_r)$ (a = half aperture).

❖(b) Parabolic dish (point focus)

- Paraboloid dish (diameter 3–25 m) with receiver at the focal point; two-axis tracking; C = 500–3000; T up to 1500 °C.
- Usually coupled with a **Stirling engine** – Dish-Stirling (25 kW), efficiency ~ 30% (highest of all solar-to-electric). Also used for solar cookers (Scheffler dish, Parabolic community cooker).

❖(c) Central receiver / Power tower (heliostat field)

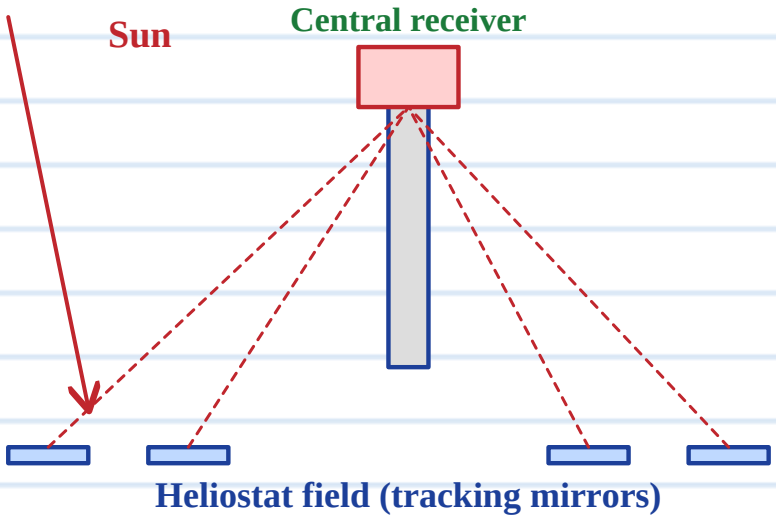


Fig 2.3 Central receiver (power tower) system

- **Heliostats** – large flat/slightly curved mirrors with two-axis tracking that all reflect sun on one **central receiver** on a tower. C = 300–1500, T = 500–1000 °C.
- Fluid: molten salt (60% $NaNO_3$ + 40% KNO_3), steam, air → storage → turbine. e.g. Solar Two, Gemasolar (Spain), Ivanpah (USA).

❖(d) Fresnel lens collector

- A thin flat plastic **Fresnel lens** (concentric grooved rings) refracts the rays to a focal point/line; low weight, low cost. Linear Fresnel reflector (LFR): long flat mirror strips on a single axis with a fixed elevated receiver.

❖(e) Compound parabolic concentrator (CPC) – Winston collector

- **Non-imaging**; two parabolic segments with different focal axes; accepts radiation over a wide acceptance angle (± 2–6°); needs **only seasonal tilt adjustment**, no daily tracking; C = 3–10; T up to 200 °C. Collects some diffuse radiation too.

8. Orientation & Tracking

- **Fixed FPC**: face south (N-hemisphere), tilt ≈ latitude (+ 10–15° for winter use). Seasonal manual adjustment gives ~ 5–10% gain.
- **Tracking modes**
  - **Polar (equatorial) mount**: one axis parallel to earth's axis, rotates 15°/h; keeps θ small.
  - **N–S horizontal axis, E–W tracking**: good in summer, lower in winter.
  - **E–W horizontal axis, N–S tracking**.
  - **Two-axis (azimuth-altitude)**: θ ≈ 0 always; best but costly.

- Trackers: mechanical (clock drive), sensor/ photo-cell based, microprocessor/ astronomical program.
- Optical errors: surface slope error, tracking error, mirror reflectivity, receiver mis-alignment → intercept factor γ.

9. Thermal Analysis of Concentrating Collectors

❖Optical efficiency

$$\eta_o = \rho \cdot \tau \cdot \alpha \cdot \gamma \text{ (reflectance × transmittance × absorptance × intercept factor)}$$

- Typical η_o = 0.70 – 0.85 for a good parabolic trough.

❖Energy balance on the receiver

- Absorbed flux $S = I_b \eta_o$ (W per m^2 of aperture, with I_b = beam radiation on aperture, including cosine factor).
- Heat loss from receiver = $U_L A_r (T_r - T_a)$.

$$Q_u = A_a S - A_r U_L (T_r - T_a) = \dot{m} C_p (T_{fo} - T_{fi})$$

- Using F_R: $Q_u = F_R [A_a S - A_r U_L (T_{fi} - T_a)]$
- Thermal efficiency:

$$\eta = Q_u / (A_a I_b) = F_R [\eta_o - U_L (T_{fi} - T_a) / (C \cdot I_b)]$$

- Key insight: the loss term is divided by the concentration ratio C → at high C, losses are small, so efficiency stays high even at high temperatures.
- Compare FPC: $\eta = F_R[(\tau\alpha) - U_L(T_{fi}-T_a)/I_T]$ (C = 1).

❖Example

> PTC: aperture 2.5 m × 10 m (A_a = 25 m^2), receiver dia 0.05 m (A_r = $\pi \times 0.05 \times 10$ = 1.57 m^2), I_b = 700, η_o = 0.75, U_L = 8, F_R = 0.9, T_{fi} = 150 °C, T_a = 30 °C.

C = 25/1.57 = 15.9. Q_u = 0.9[25×700×0.75 – 1.57×8×120] = 0.9[13125 – 1507] = **10,456 W**; η = 10456/17500 = **59.7 %**.

10. Advanced Collectors

❖(i) Evacuated tube collectors – heat-pipe type

- A sealed copper **heat pipe** with a volatile fluid (alcohol/water) inside each evacuated tube; evaporation at hot end, condensation at the manifold bulb → very fast heat transfer, can't freeze/leak. Used for hot water at 60–120 °C.

❖(ii) Compound parabolic concentrator (CPC) + evacuated tube

- Combines low heat loss with moderate concentration, no tracking.

❖(iii) Solar chimney / transpired collector

- Perforated dark metal sheet on a wall; air sucked through holes is pre-heated – efficiency 60–80% for ventilation air.

❖(iv) Photovoltaic–thermal (PV/T) hybrid collector

- A PV module with fluid tubes at the back: produces electricity **and** heat; cooling raises PV efficiency; overall efficiency ~ 60–70%.

❖(v) Integrated collector-storage system (ICS)

- The tank itself is the absorber ("batch heater") – cheap but heat loss at night.

❖(vi) Solar tracking mirror (Scheffler) / Fresnel dish

- Fixed focus, used in community kitchens (Shirdi, Mount Abu).

❖(vii) Cavity receivers and volumetric receivers

- Cavity (box) receivers trap radiation (blackbody effect) – very low re-radiation; ceramic **volumetric** receivers heat air up to 1000 °C.

❖(viii) Nanofluid / direct-absorption collectors

- Nano-particles (Al_2O_3, CuO, carbon) suspended in fluid absorb radiation directly → higher efficiency; research stage.

❖(ix) Solar tower + molten salt storage

- 24-hour dispatchable power: salts stored at 565 °C in hot tank, used on demand.

11. Comparison – FPC vs Concentrating Collector

Feature	Flat plate	Concentrating
Radiation used	Beam + diffuse	Beam only
Tracking	Not needed	Needed (except CPC)
Temperature	< 100 °C	100 – 2000 °C
Concentration ratio	1	> 1 (up to 46,000)
Cost	Low	High
Heat loss area	Large (= aperture)	Small
Maintenance	Simple	Complex (cleaning, tracking)
Uses	Hot water, heating	Power, process heat, furnaces

12. Collector Performance Terms

- **Collector efficiency** = useful energy / incident energy (instantaneous or daily).
- **Stagnation temperature:** equilibrium temperature when no fluid flows (loss = absorbed). $T_{stg} = T_a + S/U_L$.
- **Thermal capacity & response time:** time constant = $(mC)/(A\,U_L)$.
- **Incidence angle modifier** $K_{\tau\alpha} = (\tau\alpha)/(\tau\alpha)_n = 1 - b_o(1/\cos\theta - 1)$.
- **Effect of number of covers:** 1 cover – more transmittance but more loss; 2 covers used above 60 °C.
- **Effect of dust, wind & fluid flow rate** on performance.

13. Solved Problems

❖Problem 1 – Concentration ratio

A parabolic trough has aperture width 3 m and receiver diameter 6 cm. C (area basis) = 3 / (π×0.06) = 15.9.

❖Problem 2 – Maximum concentration

For a 2-D concentrator with $\theta_a = 0.267°$: $C_{max} = 1/\sin 0.267° = 1/0.00466 = 214.6$.

❖Problem 3 – Stagnation temperature

$S = 700\ W/m^2$, $U_L = 5$, $T_a = 30°C \rightarrow T_{stg} = 30 + 700/5 = 170\ °C$.

❖Problem 4 – Flow rate for given temperature rise

> Q_u = 1500 W, ΔT = 10 K, C_p = 4186 → $\dot{m}$ = 1500/(4186×10) = **0.0358 kg/s (129 kg/h).**

14. Important Questions

- Describe the construction and working of a liquid flat plate collector with neat sketch.
- Derive the Hottel-Whillier-Bliss equation; define F_R, F', U_L.
- How are concentrating collectors classified? Explain parabolic trough, dish & central receiver.
- Define concentration ratio & acceptance angle; show $C_{max} = 1/\sin\theta_a$.
- Explain CPC and its advantages. Write about thermal analysis of cylindrical parabolic collector.
- Write short notes on advanced collectors and orientation/ tracking.

> Remember: "Concentrating collectors ⇒ beam radiation only ⇒ tracking needed"

15. Solar Water Heating Systems (Application of FPC)

❖Natural circulation (thermosiphon) system

- Tank placed **above** the collector; hot water (lighter) rises into the tank, cold water descends to the collector bottom → no pump. Typical 100–200 LPD domestic system with 2 m^2 collector.
- Pressure head $\Delta p = (\rho_c - \rho_h)\, g\, H$ drives the flow.

❖Forced circulation system

- Pump + differential temperature controller; used for large systems (hospitals, hotels); tank can be placed anywhere.
- **Direct** (water in the collector) vs **indirect** (anti-freeze fluid with heat exchanger – for cold regions).

❖Design hints

- 1 m^2 of FPC heats ~ 50–60 litres/day to 60 °C; collector tilt ≈ latitude; avoid shade; insulate pipes; fit an **electric back-up** heater.
- Hard water scaling → use soft water or indirect loop.

16. Solar Cooker (Concentrating application)

- **Box type:** insulated box, glass lid, reflector mirror; 100–140 °C; slow cooking of rice/dal in 2–3 h. Efficiency ~ 25–30%.
- **Parabolic dish cooker:** reaches 250–300 °C, fast cooking, needs tracking.
- **Community/ Scheffler cooker:** steam cooking for 1000s of meals.
- Figure of merit: $F_1 = (T_p - T_a)/H_I$ (BIS test) – should be ≥ 0.12 m^2K/W.

17. Testing of Collectors (short note)

- Steady-state test: hold T_{fi} constant, measure I_T, $\dot{m}$, T_{fo} for several T_{fi} values; clear sky, near-normal incidence, wind 2–4 m/s.
- Plot η vs $(T_{fi} - T_a)/I_T$ → intercept = $F_R(\tau\alpha)$, slope = $-F_RU_L$.
- Also: time-constant test, incident-angle-modifier test, stagnation test, pressure-drop test, thermal-shock test.

18. Quick Revision Table

Term	Remember
FPC efficiency	$\eta = F_R(\tau\alpha) - F_RU_L\Delta T/I_T$
Useful heat	$Q_u = A\,F_R[S - U_L(T_{fi}-T_a)]$
Concentration ratio	$C = A_a/A_r$
C_{max} (2D / 3D)	$1/\sin\theta_a$; $1/\sin^2\theta_a$
Optical efficiency	$\rho\ \tau\ \alpha\ \gamma$
Stagnation temp	$T_a + S/U_L$

19. Worked Example – Number of Collectors for a Hostel

Need 1000 litres of hot water per day, raised from 25 °C to 60 °C. Energy = 1000×4.186×35 = **146.5 MJ/day**. Daily radiation on tilted plane = 18 MJ/m^2, average daily efficiency = 45 % → useful gain = 8.1 MJ/m^2day. Area = 146.5/8.1 = **18.1 m^2** ≈ 9 collectors of 2 m^2.

20. Sketch Tips for the Exam

- Always show the **sun rays**, label every layer (cover, air gap, absorber, insulation), mark flow direction **in / out**.

- For concentrators draw the **focal point**, aperture W, receiver and rim angle ψ_r.
- Write the governing equation beneath the diagram.